

In the late 1980s, two more production units were established in Punjab. These included the Rail Coach Factory (RCF) at Kapurthala to meet the increasing demand of coaches and the Diesel Component Works (now Diesel Loco Modernization Works) at Patiala to manufacture spare parts for diesel and electric locomotives. Manufacturing plants for wheels and axles and coil springs were set up in Bengaluru, Karnataka, and Gwalior, Madhya Pradesh. Modernization became a point of focus to inject efficiency into the production system and this was done with the help of the Central Organisation for Modernisation of Workshops (COFMOW), which was formed in 1979.

RAILWAY STAFF COLLEGE

After 1949, conversations for a central training institute gained momentum. In 1952, the Railway Staff College, housed in the Pratap Vilas Palace in Vadodara, opened. Its training methods included classroom lectures, practicals, and field visits. By the middle of the 1960s, railway training was established as a distinct discipline. The following decade saw it expanding to incorporate advanced courses, such as scientific management. The college has remodelled itself over the years in order to keep pace with changing times. Several other institutions for technical training have been established as well, such as the Training School in Pune, also known as the Indian Railways Institute of Civil Engineering, and the Indian Railways Institute of Signal Engineering and Telecommunications in modern-day Telangana.



▲ **The Railway Staff College at Vadodara** is now known as the National Academy of Indian Railways.

Setting standards and guidelines

It was only a matter of time that there would be a need for applied research in the development of railway technology. The Indian Railway Conference Association (IRCA) and the Central Standards Office (CSO) were both remnants of a colonial past. In 1952, the Research Section of the CSO was rechristened the Railway Testing Research Centre (RTRC) and assigned to provide new designs, test prototypes, and conduct research to resolve technical issues. In 1957, the CSO and RTRC merged